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HUJI Applicative Research Report — Healthcare — Week 37, 2026

3 high-potential paper(s) · Healthcare · Weekly · Week 37, 2026 · Generated September 07, 2026

Hebrew University researchers are advancing groundbreaking solutions across cancer research, from cutting-edge diagnostics to novel therapeutics. A new single-cell sequencing technology promises to revolutionize precision medicine by revealing subtle disease drivers at a lower cost and higher resolution. Additionally, two distinct therapeutic approaches target highly challenging cancers: a selective peptide for metastatic ovarian cancer and a unique antibody for immunotherapy-resistant colorectal tumors. These innovations address critical unmet needs in oncology, offering significant commercial potential for investors and industry partners.

RESEARCHERS & RESEARCH SUBJECTS — HIGH-POTENTIAL PAPERS

- **Oren Ram** — Genomics, Diagnostics, Drug Discovery, Clinical
- **Rachel Nechushtai** — Drug Discovery, Proteomics, Clinical
- **Francesca Levi-Schaffer** — Drug Discovery, Immunology, Clinical

HEALTHCARE — RESEARCH HIGHLIGHTS

1. sc-rDSeq: a robust and cost-effective full-length total RNA sequencing method for single cells reveals multilayered heterogeneity in drug-resistant lung cancer cells. 35/50

"Breakthrough single-cell sequencing dramatically lowers costs and uncovers hidden disease mechanisms for diagnostics and drug discovery."

Main researcher: Oren Ram

Department of Biological Chemistry, Alexander Silberman Institute of Life Sciences, The Hebrew University, 9190401 Jerusalem, Israel.

Published in Nucleic acids research · 2026-04

ABOUT THIS RESEARCH

Researchers have developed a new single-cell sequencing technology called sc-rDSeq that captures a much wider range of RNA types, including non-polyadenylated RNAs, at a lower cost and higher density than current standards. By applying this to lung cancer, they demonstrated the ability to identify hidden cell subpopulations and drug-resistance mechanisms that traditional methods miss.

COMMERCIAL ANGLE

The technology presents a high-value platform for next-generation precision diagnostics and drug discovery tools by providing more comprehensive transcriptomic data for identifying rare disease drivers.

COMMERCIALISATION METRICS

Novelty 8/10

The technology represents a significant platform innovation by enabling full-length, non-polyA RNA capture in a scalable droplet format, overcoming a major technical limitation of current industry standards like 10x Genomics. This capability to access the 'dark transcriptome' (ncRNA, histone RNA, etc.) provides a new dimension of biological insight that is highly relevant for precision oncology.

Commercial Potential 8/10

The technology offers a significant, platform-enabling improvement over industry standards (10x Genomics) by capturing non-polyA RNA and providing higher UMI density, which has clear commercial utility for drug discovery and precision oncology. Its scalability and cost-effectiveness position it as a highly licensable tool for large-scale clinical biomarker profiling.

Market Size 8/10

The technology targets the high-value single-cell sequencing market by offering a platform-enabling solution for comprehensive transcriptomics, which is essential for advanced drug discovery and personalized oncology. Its ability to capture non-polyA RNAs and alternative splicing at scale provides a significant competitive edge in the expanding precision medicine and biomarker discovery sectors.

Tech Readiness 4/10

The technology is currently at the proof-of-concept stage, demonstrated through successful validation in a laboratory setting using cancer cell lines. While it shows significant performance improvements over existing methods, it has not yet been transitioned into a standardized, commercially available kit or integrated into clinical diagnostic workflows.

IP Strength 7/10

The technology represents a potentially patentable platform innovation by optimizing droplet-based capture for non-polyA RNA, offering a clear technical advantage over industry standards like 10x Genomics. However, defensibility may be challenged by the crowded landscape of single-cell sequencing chemistries and the difficulty of protecting specific microfluidic or bead-based optimizations once widely adopted.

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2. Targeting primary and metastatic ovarian cancer with a peptide derived from the human NAF-1/CISD2 protein. 31/50

"A novel peptide therapy selectively eradicates primary and metastatic ovarian cancer cells with low toxicity."

Main researcher: Rachel Nechushtai · rachel@mail.huji.ac.il.

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Published in Biomedicine & pharmacotherapy = Biomedecine & pharmacotherapie · 2026-07

ABOUT THIS RESEARCH

Researchers have developed a specialized peptide derived from the NAF-1/CISD2 protein that selectively destroys primary and metastatic ovarian cancer cells. The treatment showed efficacy and low toxicity in both laboratory settings and human ovarian cancer mouse models.

COMMERCIAL ANGLE

Development of a first-in-class targeted peptide therapeutic to address the high relapse rates and low survival outcomes of advanced ovarian cancer.

COMMERCIALISATION METRICS

Novelty 6/10

The use of a protein-derived peptide for selective cancer cytotoxicity is a known strategy; however, targeting the NAF-1/CISD2 pathway specifically for ovarian cancer represents a novel application of this protein's biology.

Commercial Potential 7/10

The peptide addresses a high-unmet-need market (refractory ovarian cancer) with demonstrated selectivity and in vivo efficacy, offering a clear path to a therapeutic product. However, the score is tempered by the typical pharmacological challenges of peptides, such as stability and delivery, which are not addressed in the abstract.

Market Size 8/10

Ovarian cancer represents a high-value target market due to a massive global incidence rate and a critical lack of effective treatments for relapsed or metastatic cases. The addressable demand is significant given the high relapse rate (80%) and poor 5-year survival outcomes.

Tech Readiness 3/10

The technology is at a basic proof-of-concept stage, having demonstrated efficacy in vitro and in xenograft mouse models, which corresponds to early preclinical development.

IP Strength 7/10

The innovation centers on a specific peptide sequence derived from a human protein (NAF-1/CISD2), which is typically highly patentable as a composition of matter. While the specificity for ovarian cancer provides strong defensibility, the score is tempered by the inherent vulnerability of peptides to proteolysis and potential ease of sequence modification by competitors.

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3. The anti-CEACAM1 agonistic antibody CCM5.01 inhibits the growth of colon cancer cell lines in 2D and 3D models. 30/50

"New antibody immunotherapy targets drug-resistant colorectal cancer, opening avenues for previously untreatable tumors."

Main researcher: Francesca Levi-Schaffer · francescal@ekmd.huji.ac.il.

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Published · 2026-08-31

ABOUT THIS RESEARCH

Researchers have demonstrated that a specific monoclonal antibody, CCM5.01, can inhibit the growth of colorectal cancer cells by targeting the CEACAM1 receptor. The study shows that the antibody's effectiveness is linked to the ratio of long to short isoforms of the receptor, particularly in resistant tumor types.

COMMERCIAL ANGLE

Development of a novel immunotherapy targeting the CEACAM1 pathway to treat microsatellite-stable (MSS) colorectal cancers that are currently resistant to standard checkpoint inhibitors.

COMMERCIALISATION METRICS

Novelty 5/10

The research provides important validation of an existing lead candidate (CCM5.01) in a new disease indication (CRC) rather than introducing a novel mechanism or a new class of molecule. While the link between the L/S isoform ratio and efficacy is valuable for precision medicine, the core innovation (the agonistic antibody) was already established in prior work.

Commercial Potential 7/10

The work identifies a specific, actionable therapeutic target (CEACAM1) and a lead candidate (CCM5.01) for the high-value MSS colorectal cancer market, which is currently underserved by existing immunotherapies. However, the commercial potential is tempered by the fact that the data is limited to in vitro and 3D models, lacking the in vivo efficacy data required to validate a robust drug development program.

Market Size 9/10

The target addresses the massive and high-unmet-need colorectal cancer market, specifically the large microsatellite-stable (MSS) patient population that is currently resistant to standard immunotherapies. By targeting CEACAM1, this approach has the potential to unlock a multi-billion dollar market segment within oncology.

Tech Readiness 3/10

The research is currently in the preclinical, in vitro stage, focusing on cell line proliferation and 3D spheroid models. It lacks in vivo animal data or pharmacokinetic studies necessary to move toward clinical candidate status.

IP Strength 6/10

The innovation centers on a specific monoclonal antibody (CCM5.01) targeting a validated oncology target, which is highly patentable; however, the abstract focuses on mechanistic validation in cell lines rather than demonstrating a novel platform or proprietary structural modifications that would ensure high defensibility against future competitors.

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